

# Siddharth Gautam

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## SUMMARY

Electronics and Communication Engineering undergraduate at NSUT Delhi with hands-on experience in RTL design, digital verification, and FPGA prototyping. Proficient in Verilog/SystemVerilog with demonstrated projects spanning processor microarchitecture, communication protocols, and embedded systems. Seeking opportunities in VLSI design, hardware verification, or embedded systems engineering.

## EDUCATION

**Netaji Subhas University of Technology (NSUT), Delhi**

Delhi, India

*B.Tech – Electronics and Communication Engineering*

*2023 – 2027*

## EXPERIENCE

**Research Intern**

May 2025 – Jul 2025

*NSIT, Delhi*

*Delhi, India*

- Conducted computational modeling of  $\text{Cs}_3\text{Bi}_2\text{I}_9$  perovskite as a non-toxic alternative for sustainable thin-film photovoltaic applications using SCAPS-1D simulation software.
- Optimized device architectures by systematically tuning absorber layer thickness (200–600 nm) and carrier concentrations, achieving a simulated power conversion efficiency improvement of over 15%.

## PROJECTS

**RISC-V 5-Stage Pipelined Processor** | *Verilog, Xilinx Vivado*

🔗 Source Code

- Designed and implemented a 32-bit RISC-V (RV32I) CPU in **Verilog** with a 5-stage pipeline (IF/ID/EX/MEM/WB), full hazard detection unit, and data-forwarding paths to eliminate pipeline stalls.
- Achieved a post-synthesis clock frequency of **96 MHz** on Xilinx Artix-7 FPGA by reducing critical-path delay; validated correctness across 50+ test programs via GTKWave simulation.

**SPI Protocol Controller** | *System Verilog, GTKWave*

🔗 Source Code

- Developed a parameterized **SPI Master-Slave** controller in SystemVerilog supporting all four SPI modes (CPOL/CPHA), configurable clock dividers, and 8-to-32-bit frame widths.
- Achieved **100% functional coverage** using SystemVerilog Assertions (SVA) and a constrained-random UVM-style testbench environment, catching 3 RTL bugs before tape-out simulation.

**UART Transmitter & Receiver** | *Verilog, FPGA*

🔗 Source Code

- Implemented a fully synthesizable UART core with programmable baud rate generation (9600–115200), configurable parity (odd/even/none), and **FWFT FIFO** buffering for burst traffic handling.
- Reduced LUT utilization by **22%** and Flip-Flop usage by **18%** versus a baseline implementation through RTL micro-optimizations, enabling deployment on resource-constrained Spartan-7 devices.

**DIY Prosthetic Hand Control System** | *Embedded C++, Arduino*

🔗 Source Code

- Designed a real-time embedded control system using **Muscle Bio-Amp patchy** sensors and Arduino Nano to acquire sEMG signals and translate muscle activation patterns into servo motor movements.
- Developed adaptive thresholding algorithms in **Embedded C++** achieving end-to-end actuation latency under **50 ms**, with a modular software architecture enabling easy addition of new gesture profiles.

## TECHNICAL SKILLS

**HDL / Languages:** Verilog, SystemVerilog, C/C++, Python (scripting & automation), Shell Scripting, Assembly (RISC-V)

**EDA Tools:** Xilinx Vivado, Cadence Virtuoso, GTKWave, ModelSim

**Verification:** SystemVerilog Assertions (SVA), Constrained-Random Simulation, Functional Coverage

**Hardware & Protocols:** RTL Design, Digital Logic Synthesis, FPGA Prototyping, SPI, UART, I2C

**Core Subjects:** Digital System Design, Computer Architecture, VLSI Design, Embedded Systems

**Other:** Git/GitHub, Linux/Bash, Technical Documentation